

2nd Feb 2026 / DA6360

⇒ 1 Dimensional Optimization Problems (unconstrained)

- behaviour of $f(x)$
- x^* look like (stopping criterion)
- convergence of sequences

$z_1 \dots z_n$ (seq. of real nos.)
when does this seq. converge?

$z_1 \dots z_n \rightarrow z^*$
→ as we get closer as $n \rightarrow \infty$
how to define this closeness?
this is arbitrary $\epsilon > 0$

For any $\epsilon > 0$, we must ^{eventually} show that z_1 & z_2 are closer than ϵ . (stay there)

- For any $\epsilon > 0$, there exists n_ϵ s.t.

arbitrary $|z_n - z^*| < \epsilon$, $\forall n \geq n_\epsilon$
↓ this is defined as closeness
eventually

eg. $z_n = 1 \pm (-1)^n 0.5$

1.5	-----
0.5	-----

$z_n = 1 + \frac{(-1)^n 0.5}{n}$

- Continuity $\rightarrow$ A funcⁿ is said to be continuous at $\hat{x}$ if
- $\rightarrow$ there is no jump (arbitrary) $\epsilon > 0$
 - $\rightarrow$ do not lift pen
 - $\rightarrow$ approach from both ends, we should meet at the same pt.

$$\boxed{x_n \rightarrow \hat{x} \Rightarrow f(x_n) \rightarrow f(\hat{x})}$$

- we have $\epsilon > 0$ then $\delta_\epsilon > 0$ s.t. $\forall$

$|x - \hat{x}| < \delta_\epsilon$, it is true that $|f(x) - f(\hat{x})| < \epsilon$ (this defines the no jump statement)

$\forall \delta \in (0, \delta_\epsilon)$, $|x - \hat{x}| < \delta$

Basically it says that I am making some Δ , but that Δ is within a certain threshold, then, I have not jumped.

- Jump discontinuity

$$f(x) = \begin{cases} 0 & x < 1 \\ 1 & x > 1 \end{cases}$$

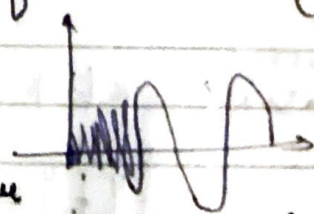


Essential

Discontinuity

$$f(x) = \sin\left(\frac{1}{x}\right)$$

no jump, but have to be very fast (oscillate)



- Differentiability - A funcⁿ is differentiable at $\hat{x}$ if -

$$\left[f'(x) = \lim_{x \rightarrow \hat{x}} \frac{f(x) - f(\hat{x})}{x - \hat{x}} \text{ exists} \right]$$

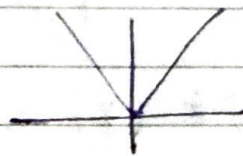
whichever seq. you reach

$$\left\{ \frac{f(x_i) - f(\hat{x})}{x_i - \hat{x}} \right\}_{i=1}^{\infty} \rightarrow f'(\hat{x})$$

eg. $f(x) = x^2$



cont. ✓
diff. ✓



cont. ✓
diff. ✗

differentiable funcⁿ in $[a, b]$

$$\int_a^b f'(x) dx = f(b) - f(a)$$

$$f(b) = f(a) + \int_a^b f'(x) dx$$

Laurens
Motors

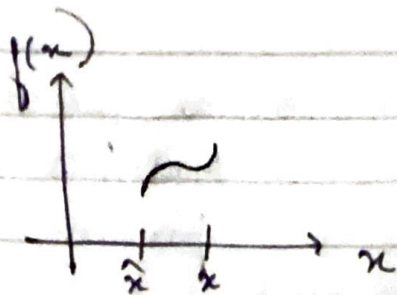
$$x(t) = x(0) + \int_0^t x'(\tau) d\tau$$

$$x''(t) = x'(0) + \int_0^t x''(\tau) d\tau \quad \left| \begin{array}{l} x=0 \rightarrow \text{body at rest} \\ x'=t \rightarrow \text{body at const. motion} \end{array} \right.$$

if that cause x'' if for
1 obj. then n times causes the same for
 n obj.

$$F = ma$$

First order approx. at $\hat{x}$ Summation over the loop variable.



$$f(x) = f(\hat{x}) + \int_{\hat{x}}^x f'(\tau) d\tau$$

$$\sum_{i=1}^n \dots$$

$$f(x) = f(\hat{x}) + \int_{\hat{x}}^x f'(\tau) d\tau$$

$$= f(\hat{x}) + f'(\hat{x}) [\tau]_{\hat{x}}^x$$

$$= f(\hat{x}) + f'(\hat{x}) [x - \hat{x}]$$

- Second Order Approximation at $\hat{x}$

$$f(x) = f(\hat{x}) + \int_{\hat{x}}^x f'(\tau) d\tau$$

$$= \left(f(\hat{x}) + \int_{\hat{x}}^x (f'(\hat{x}) + \int_{\hat{x}}^z f''(y) dy) d\tau \right)$$

$$= f(\hat{x}) + \int_{\hat{x}}^x f'(\hat{x}) d\tau + \int_{\hat{x}}^x \int_{\hat{x}}^z f''(\hat{x}) dy d\tau$$

$$= f(\hat{x}) + f'(\hat{x}) (x - \hat{x}) + \int_{\hat{x}}^x (\tau - \hat{x}) d\tau$$

$$= f(\hat{x}) + f'(\hat{x}) (x - \hat{x}) + f''(\hat{x}) \left[\frac{(\tau - \hat{x})^2}{2} \right]_{\hat{x}}^x$$

$$= f(\hat{x}) + f'(\hat{x}) (x - \hat{x}) + f''(\hat{x}) \left(\frac{(x - \hat{x})^2}{2} \right)$$

- it gets higher, then it gets better.